VibeMatch

A Project Report Submitted

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by

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ABSTRACT

VibeMatch is a mood-based social discovery web application designed to help users find activities, events, and people based on how they feel at a given moment. Instead of manually searching for events, users can simply select their current mood—such as chill, party, study, or food hunt and the platform suggests activities that match their vibe.

The application provides features such as event browsing, event creation, mood-based filtering, map-based discovery, and interactive chat functionality for event participants. The interface is designed to be modern, responsive, and easy to navigate.

The front-end of the application is developed using React, which allows the interface to be built using reusable components and dynamic rendering. Bootstrap / ReactStrap is used to create responsive layouts and styled UI components. These technologies together enable the development of a scalable and user-friendly web interface.

The project demonstrates the use of modern frontend development practices such as component-based architecture, state management, responsive design, and modular UI development.

Table of Contents

ABSTRACT 2
1. HIGH LEVEL DESIGN 4
1.1 System Architecture 4
..... 4
1.2 Description 4
2. COMPONENTS/CONCEPTS USED 6
3. SCREENSHOTS AND DESCRIPTIONS 7
4. DETAILS OF INDIVIDUAL CONTRIBUTION 10
5. CONCLUSION 11

1. HIGH LEVEL DESIGN

1.1 System Architecture

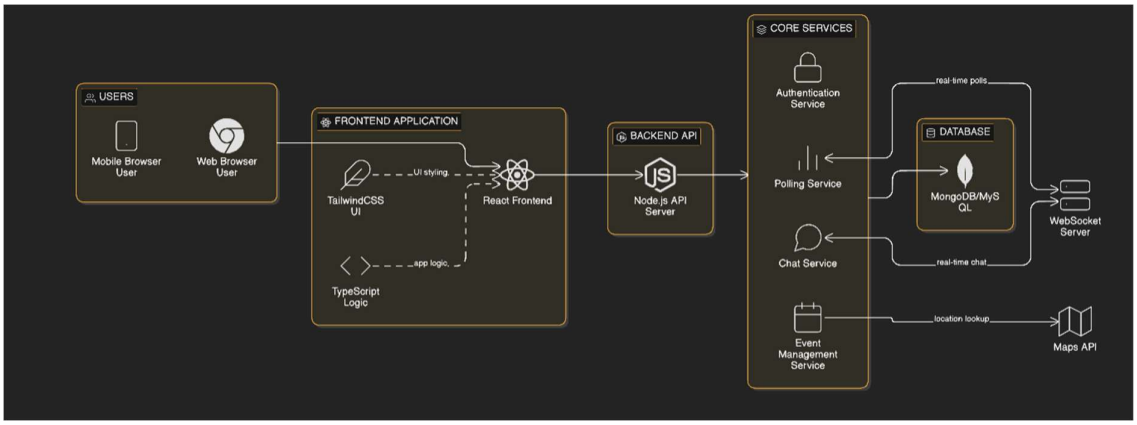

Figure 1: System Architecture Diagram

1.2 Description

The VibeMatch frontend is structured using a component-based architecture provided by React.

The user interacts with the application through a web browser. The React application renders different pages and components based on user actions such as selecting moods, browsing events, or creating new events.

The system consists of several major components:

- **Landing Page**

Introduces the platform and allows users to explore the application.

- **Dashboard**
  Displays mood selection options and recommended events.

- **Event Management Interface**
  Allows users to create and join events.

- **Map Discovery Interface**
  Shows events on a location-based map.

- **Chat Interface**
  Allows event participants to communicate with each other.

- **User Profile Page**

Displays user details, events, and activity history.

React components communicate with each other through props and state, ensuring efficient UI updates.

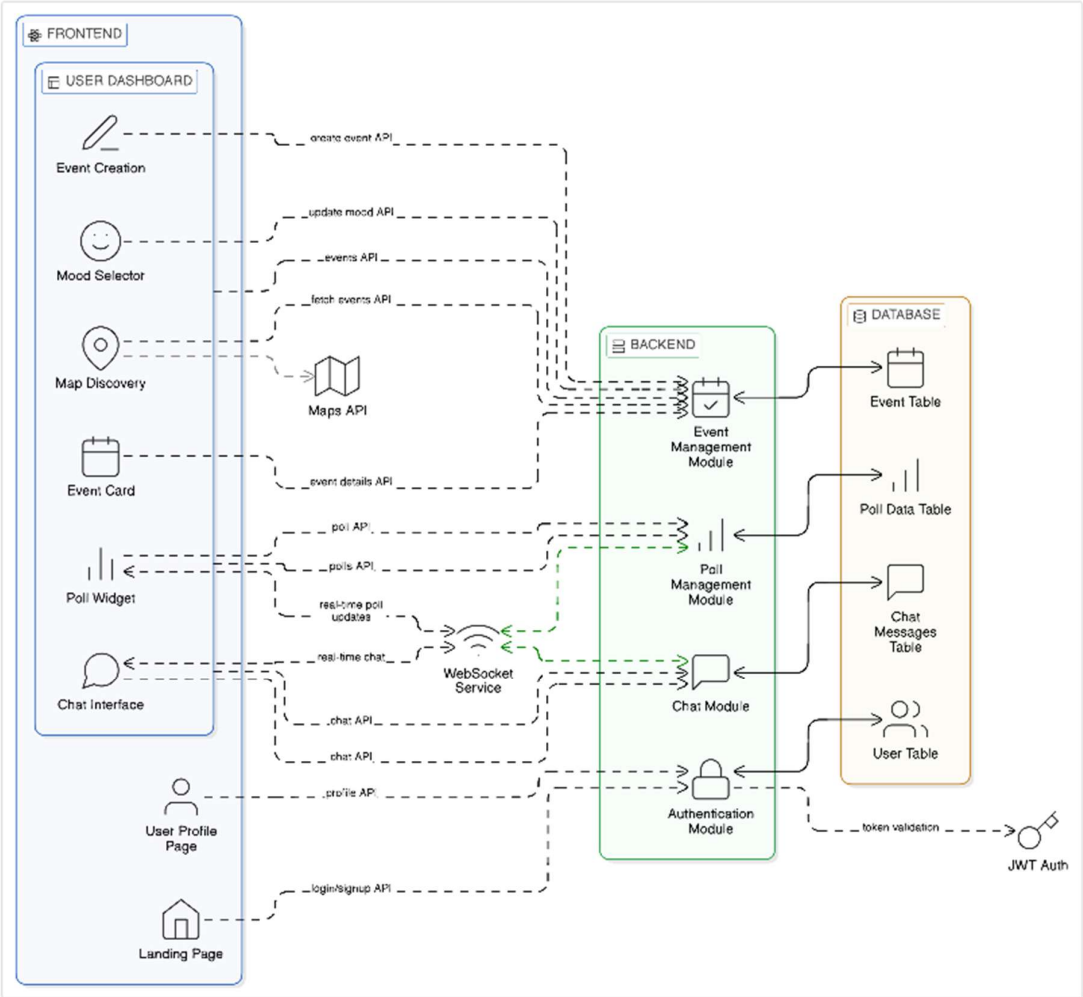

Figure 2: Component Architecture Diagram

2. COMPONENTS/CONCEPTS USED

Component / Concept	**Technology Used**	**Purpose**
React Framework	React.js	Used to build the interactive frontend application
UI Framework	Bootstrap / ReactStrap	Provides responsive layouts and styled components
Component-Based Architecture	React Components	Enables reusable UI elements
Routing	React Router	Manages navigation between pages
State Management	React Hooks	Handles dynamic data and UI updates
Mood Selector	Custom React Component	Allows users to select their current mood
Event Cards	Custom Component	Displays event details
Forms	Bootstrap Forms	Used for event creation and user inputs
Responsive Layout	Bootstrap Grid System	Ensures the application works across different screen sizes

3. SCREENSHOTS AND DESCRIPTIONS

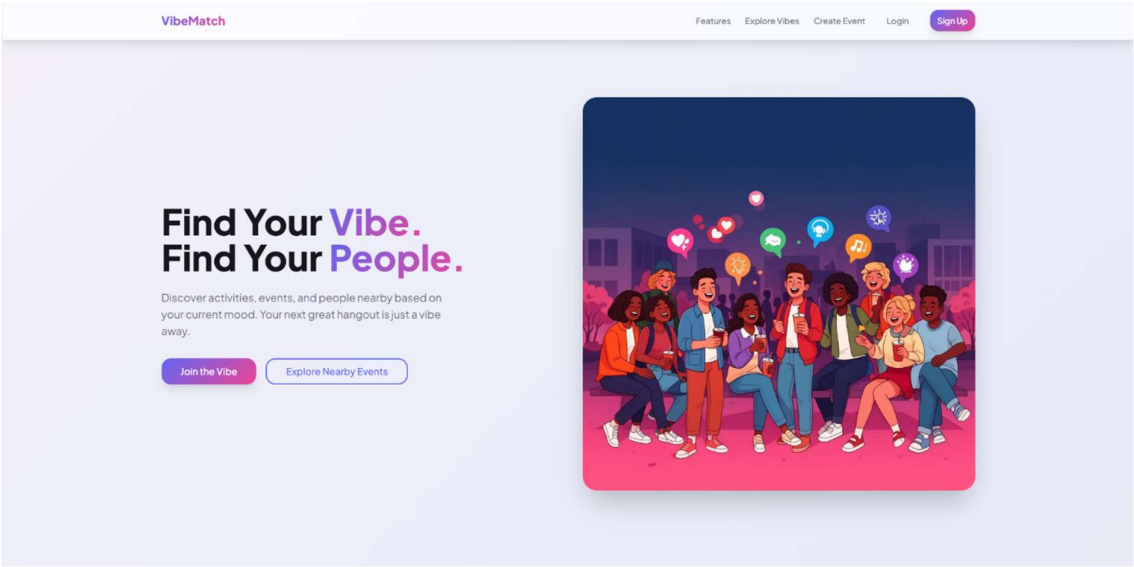

Figure 3: Homepage

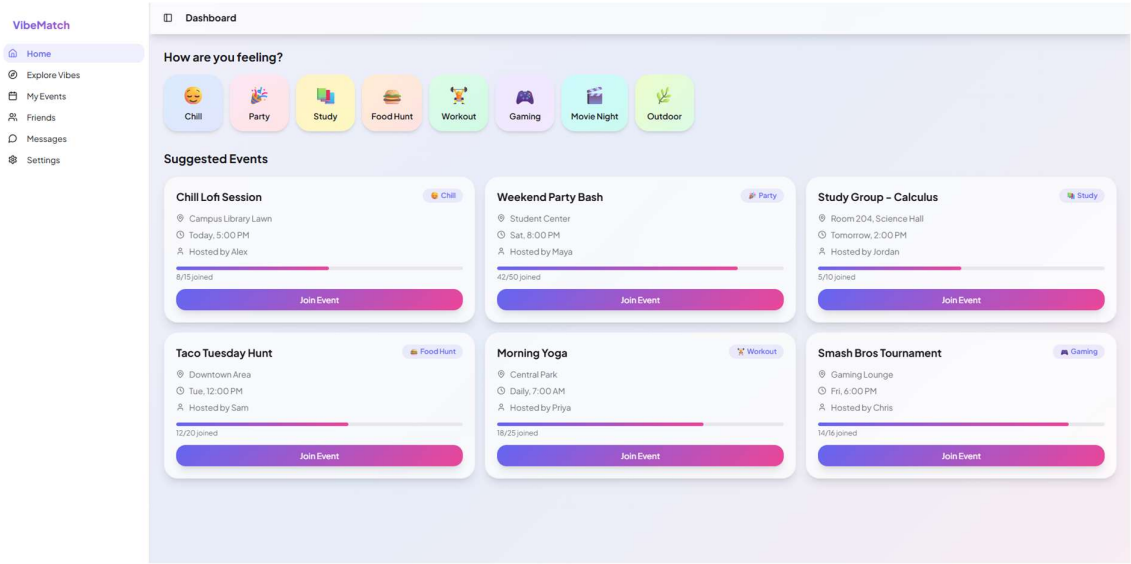

Figure 4: The dashboard allows users to select their current mood and discover events based on that mood.

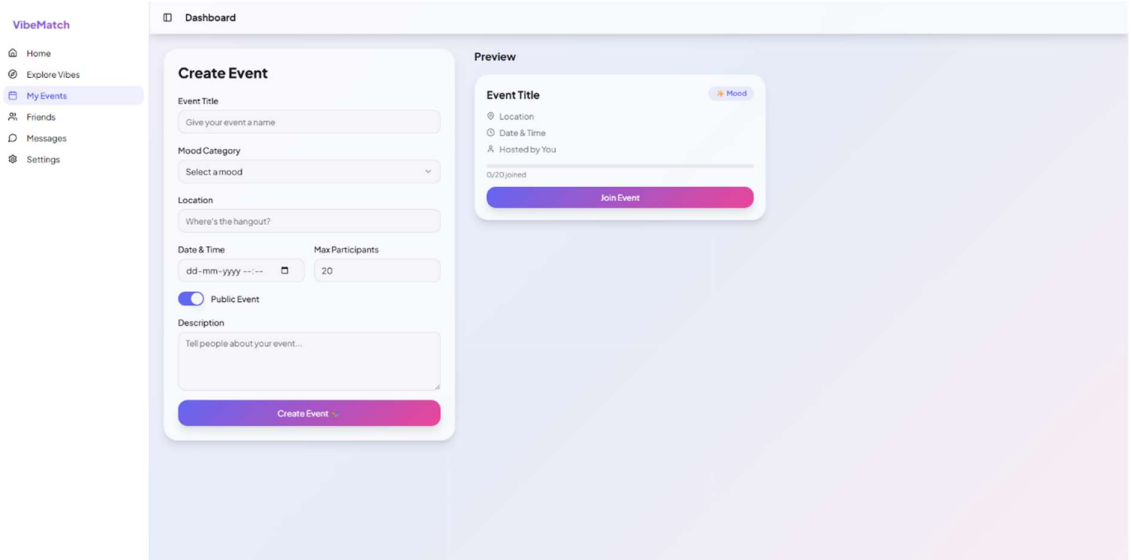

Figure 5: Users can create events by entering information such as event title, location, date, and mood category.

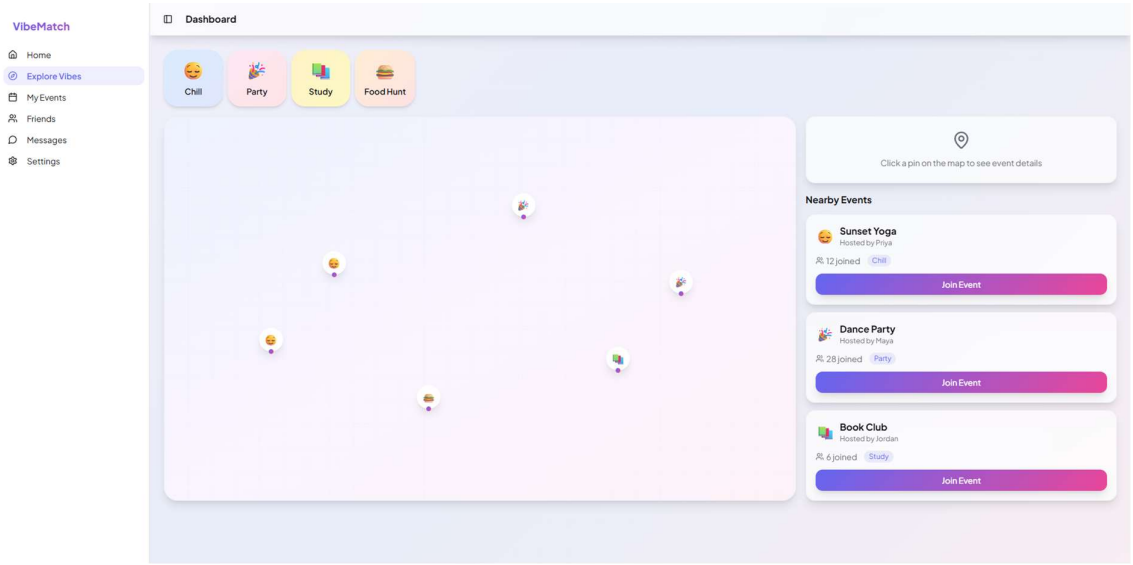

Figure 6: The map interface allows users to visually explore nearby events and join activities that interest them

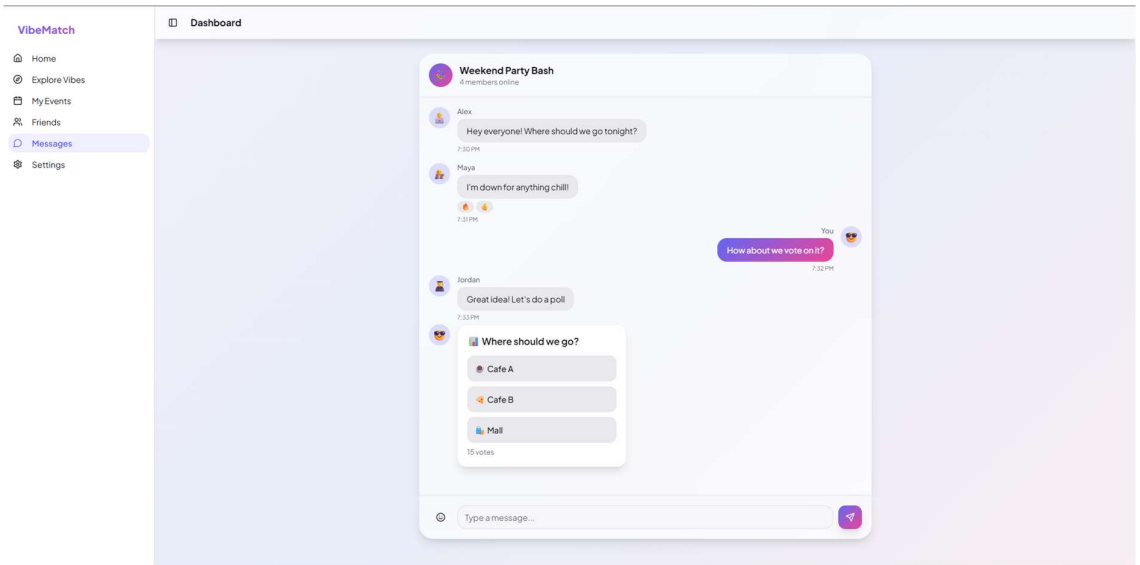

Figure 7: Participants can communicate in real time and coordinate event plans through the chat interface.

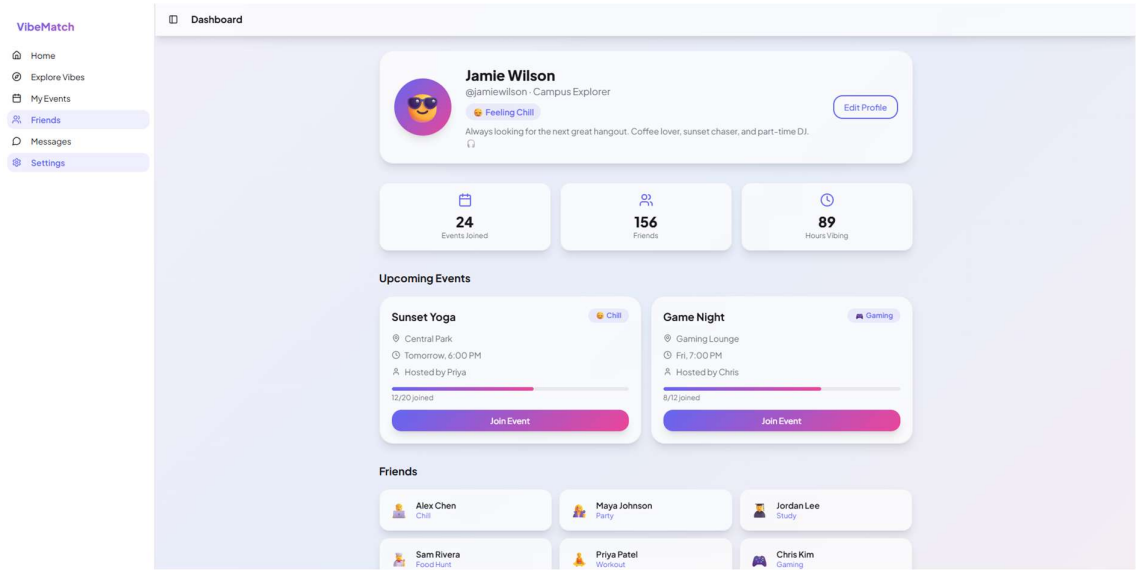

Figure 8: The profile page displays user information, upcoming events, and social activity within the platform.

4. DETAILS OF INDIVIDUAL CONTRIBUTION

Team Member	Contribution
Suniket Sen	Dashboard implementation, event creation interface and event display components
Ishan Satyanand Thakur	Map discovery interface, chat UI components, and feature integration
Vedant Agarwal	Frontend UI design and React component development,

5. CONCLUSION

The VibeMatch project demonstrates how modern frontend technologies can be used to develop an interactive and user-friendly social discovery platform. By using React and Bootstrap, the application provides a responsive and modular interface that allows users to explore events, select moods, and interact with other participants.

The component-based architecture ensures that the application is scalable and maintainable. Through this project, the team gained practical experience in React development, UI design, component-based programming, and responsive web design.

Future improvements could include backend integration, real-time data updates, mobile application support, and AI-based event recommendations to further enhance the platform's capabilities.